SHAP-σ: A Label-Free Rashomon Pipeline for Multivariate Time-Series Anomaly Detection

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***Abstract*—We address unsupervised anomaly detection on multivariate cellular network KPI streams, where labels are unavailable for both training and model selection. Existing label-free quality criteria such as the Mass–Volume curve [1] are known to be noisy, and single-detector SHAP explanations cannot signal when a flag should be trusted. We propose SHAP-σ, a label-free pipeline for per-KPI multivariate anomaly detection that (i) self-tunes each detector's hyperparameters without labels (PCA via 95% explained variance, k-NN/LOF via k-distance elbow, GMM via BIC, HBOS via Sturges' rule); (ii) selects a small *Rashomon set* via a closed-form joint pair score combining a Mass–Volume proxy with a cardinality veto; (iii) explains every selected detector with a LightGBM surrogate and TreeSHAP; and (iv) uses the cross-detector standard deviation of those attributions as a per-feature gate that corroborates anchor flags. On TELCO (12 per-KPI label streams), the closed-form pair score at $K_R{=}2$ identifies the empirical oracle pair (ECOD+HBOS) label-free, and the pipeline raises per-KPI macro F_1 from 0.401 (best singleton) to 0.440 ($+0.039$, isolated $\Delta\sigma{=}+0.038$), with precision rising from 0.488 to 0.641 ($+31\%$ relative). SHAP-σ exceeds released deep baselines DC-VAE ($F_1{=}0.134$) and C-MTAD-GAT ($F_1{=}0.160$) by 2.8-$3.3\times$ on the same evaluation window.**

***Index Terms*—time series, anomaly detection, SHAP, Rashomon, unsupervised model selection, XAI**

I. Introduction

Unsupervised anomaly detection in multivariate time series underpins fault diagnosis in domains as diverse as cellular networks, server fleets, and industrial telemetry. Yet a persistent obstacle hinders deployment: without ground-truth labels, the practitioner cannot reliably decide which detector to trust. Internal validation metrics that work well for clustering rarely transfer to anomaly detection, and recent large-scale empirical work shows that most label-free criteria perform near random when ranking modern detectors [2].

This selection problem is sharpened by the *Rashomon effect* [3]: for any non-trivial task, a multiplicity of near-optimal models exists, each making different errors [4], [5]. We argue that this attribution disagreement is not noise to be averaged away but a signal to be exploited: at a flagged point, the spread of attributions across the Rashomon set indicates that at least one detector commits to a feature-level explanation, providing second-source evidence beyond the anchor's score.

Prior label-free model selection has explored three avenues. Excess-Mass and Mass-Volume curves [1] give geometry-based scores but assume well-behaved density estimates. Meta-learning approaches such as MetaOD [6] and ELECT [7] transfer historical-performance signals across datasets, but require an offline labelled meta-corpus. Most recently, robust Borda rank aggregation across multiple imperfect internal metrics has been proposed for time-series AD [8]; the large-scale study of [2] shows that no internal metric universally dominates. PyOD [9] has standardized detector implementations. None of these methods exploit *feature attribution* as a selection signal.

Independently, SHAP [10], [11] has become the de facto post-hoc explainer for anomaly detection, mostly to explain a single chosen model; the closest precedent is Levy et al. [12], who use SHAP attributions for ensemble member *selection* in supervised tabular AD, whereas we use cross-detector SHAP standard deviation as an attribution-evidence *gate* in the unsupervised TS regime. Modern deep TS-AD methods [13]–[16] likewise commit to a single architecture. Outlier ensemble theory [17], [18] and the classical diversity literature [19] establish that complementary errors improve ensemble accuracy, yet the standard diversity measures (Q-statistic, disagreement) require labels. Following [18], we replace label-dependent diversity with the Spearman rank correlation of training scores, which [18] identifies as the practical label-free proxy in OD ensembles.

a) Contributions.: (1) Label-free self-tuning of the entire PyOD detector pool (PCA via 95% explained variance; k-NN/LOF via k-distance elbow; GMM via BIC; HBOS via Sturges' rule), which raises detector-level F_1 without using any anomaly labels. (2) A closed-form Rashomon pair score $s_{ij} = \frac{1}{2}(q_i + q_j)(1 - \rho_{ij})$, where q_d combines a Mass–Volume proxy with a cardinality veto (eliminating constant-output detectors), jointly rewarding label-free quality and inter-detector score decorrelation. (3) A σ-SHAP attribution-evidence filter using cross-detector SHAP standard deviation as a second-source signal that corroborates anchor flags on per-KPI labels. (4) Empirical demonstration on the TELCO cellular benchmark that the closed-form pair score identifies the empirical oracle pair label-free, and the pipeline exceeds the best singleton and recent deep baselines DC-VAE and C-MTAD-GAT [20], [21].

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Code and data for full reproducibility: `https://github.com/ugurdar/shap_sigma_paper`.

b) Literature gap.: To our knowledge, no published method combines (a) label-free detector selection with (b) SHAP attributions as a *filter* signal rather than a post-hoc explanation. Rashomon-set analysis has so far been developed in supervised regimes [4], [5]; its extension to unsupervised anomaly detection is open. Robust rank aggregation [8] pools imperfect label-free metrics but never inspects attributions, while [2] diagnoses their weakness without proposing a disagreement-based pipeline. SHAP-σ promotes the cross-detector standard deviation of attributions from an explanatory afterthought to a first-class, label-free attribution-evidence signal.

The remainder of the paper is organized as follows. Section II formalizes joint pair scoring and the σ-SHAP filter. Section III describes datasets and protocol. Section IV reports empirical results. Section V concludes.

II. Methodology

We propose **SHAP-σ**, a label-free pipeline that selects a small subset $\mathcal{R} \subseteq \mathcal{D}$ of detectors from a pool $\mathcal{D}$ by jointly optimising label-free quality and pairwise diversity, explains every member of $\mathcal{R}$ with a tree surrogate, takes TreeSHAP attributions [10], [11], and converts the cross-detector spread of those attributions into a per-point, per-feature attribution-evidence gate. Labels are used nowhere in training, ranking, or set selection.

A. Notation

Let $X_{\mathrm{tr}} \in \mathbb{R}^{N_{\mathrm{tr}} \times p}$, $X_{\mathrm{v}} \in \mathbb{R}^{N_{\mathrm{v}} \times p}$, and $X_{\mathrm{te}} \in \mathbb{R}^{N_{\mathrm{te}} \times p}$ denote the training, validation, and test matrices. The detector pool is $\mathcal{D} = \{d_1, \ldots, d_K\}$ with $K{=}8$. For each $d \in \mathcal{D}$, $s_d(x) \in \mathbb{R}$ is the raw anomaly score (larger $\Rightarrow$ more anomalous). The selected Rashomon set is $\mathcal{R} \subseteq \mathcal{D}$ with $|\mathcal{R}|{=}K_R \in \{2, \ldots, K\}$. We write $a \in \mathcal{R}$ for the anchor.

B. Detector Pool and Label-Free Self-Tuning

$\mathcal{D}$ consists of $K{=}8$ heterogeneous PyOD detectors [9]: Isolation Forest [22], LOF [23], ECOD [24], COPOD [25], k-NN [26], PCA [27], a Gaussian Mixture density score [17], and HBOS [28]. The pool deliberately mixes tree-, density-, distance-, projection-, and histogram-based inductive biases, so that the cross-detector σ in Stage 4 is informative.

a) Self-tuning.: PyOD default hyperparameters often fail silently – e.g. PCA with `n_components`=None collapses to constant-zero scores on some inputs. We therefore self-tune every parameterised detector *label-free* from X_{tr} alone:

- PCA: `n_components` chosen so that cumulative explained variance is $\geq 95\%$;
- k-NN and LOF: `n_neighbors` from the k-distance elbow (the rank that maximises the deviation of the mean k-th-nearest-neighbour distance from the diagonal of its normalised cumulative curve);
- GMM: `n_components` that minimises the Bayesian Information Criterion (BIC) on X_{tr} over $K \in \{1, \ldots, 10\}$;
- HBOS: `n_bins` from Sturges' rule, $\lfloor \log_2 N_{\mathrm{tr}} \rfloor + 1$;
- IF, ECOD, COPOD: defaults (no informative label-free signal available for IF; ECOD and COPOD are parameter-free).

On TELCO this self-tuning leaves the headline detector ECOD unchanged but markedly improves precision for the k-NN/LOF/PCA pair, which is what the joint pair score and σ-filter consume downstream.

C. Stage 1 – Label-Free Quality via a Mass–Volume Proxy

We need a continuous label-free quality score $q_d \in [0, 1]$. Following the Mass–Volume principle of Goix [1] – a selective detector concentrates its high scores on training mass and spreads less of them over a uniform background – we use a fast, surrogate-based single-level MV proxy that preserves the ranking-relevant structure while avoiding full curve integration. For each d, we fit a LightGBM surrogate $\hat{s}_d$ on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$ and define

$$A_d \;=\; \Pr_{X_{\mathrm{u}}}\big[\hat{s}_d(X_{\mathrm{u}}) \;>\; Q_{0.9}\big(s_d(X_{\mathrm{tr}})\big)\big], \tag{1}$$

the surrogate-predicted background tail mass above the empirical 0.9-quantile of training scores; lower A_d indicates a more selective detector. We then min–max invert,

$$q_d \;=\; \frac{\max_e A_e - A_d}{\max_e A_e - \min_e A_e} \;\in\; [0, 1], \tag{2}$$

estimating Eq. (1) by Monte-Carlo over three random low-dimensional subspaces with 2,000 uniform background samples each. We use a single-level MV proxy rather than the full EM/MV average [1]: [2] reports that combining imperfect internal metrics need not improve ranking, and on TELCO EM-only elects PCA (per-KPI $F_1{=}0.04$) while Eq. (1) elects ECOD ($F_1{=}0.40$). A full Goix EM/MV-AUC ablation is deferred to a journal extension.

D. Stage 2 – Joint Pair Scoring and the Rashomon Set

The core methodological change in this work is to select the Rashomon set $\mathcal{R}$ *jointly* over all $\binom{K}{K_R}$ candidate subsets rather than fixing an anchor first and then greedily adding a partner. Joint selection makes the pipeline a Rashomon procedure in the sense of [4], [5]; the closest supervised precedent is the selective-ensemble construction of Black et al. [29], which we adapt to the unsupervised setting by replacing label-dependent agreement with Spearman score correlation.

a) Pairwise diversity.: Let $\rho_{ij} = \mathrm{Spearman}(s_i(X_{\mathrm{tr}}), s_j(X_{\mathrm{tr}}))$. Diversity is $1 - \rho_{ij} \in [0, 2]$; rank correlation is robust to the heterogeneous score scales of $\mathcal{D}$ [19].

b) Three label-free strategies.: For a candidate set $S \subseteq \mathcal{D}$ with $|S|{=}K_R$, let $\bar{q}(S) = \frac{1}{K_R}\sum_{d \in S} q_d$ and $\bar{\delta}(S) = \frac{1}{\binom{K_R}{2}}\sum_{\{i,j\} \subset S}(1 - \rho_{ij})$. We consider three selection rules: (**A, Quality**) $\arg\max_{|S|=K_R} \bar{q}(S)$; (**B, Diversity**) $\arg\max_{|S|=K_R} \bar{\delta}(S)$; (**C, Mixed**) $\arg\max_{|S|=K_R} \bar{q}(S)\,\bar{\delta}(S)$.

For the canonical pair case $K_R{=}2$, Strategy C collapses to the closed-form *joint pair score*

$$(i^*, j^*) \;=\; \arg\max_{1 \leq i < j \leq K} \frac{q_i + q_j}{2}\big(1 - \rho_{ij}\big), \tag{3}$$

which is the centrepiece of our method.

c) Anchor.: Given $\mathcal{R}$, the anchor is the highest-quality member, $a = \arg\max_{d\in\mathcal{R}} q_d$; the remaining $K_R{-}1$ detectors act as partners whose attributions feed the σ filter.

d) Default and dataset-dependence.: The default pipeline is $(K_R, \mathcal{S}){=}(2, \text{Mixed})$; Eq. (3) then gives a single closed-form selection rule. Empirically (Sec. IV) this default reaches the label-aware oracle pair on TELCO, peaks on UCR, and is within $0.01\,F_1$ of the per-dataset best on SMD (where $(5, \text{Quality})$ is marginally better). We expose $(K_R, \mathcal{S})$ as label-free hyperparameters following [2], [8], who argue that no internal criterion universally dominates; our contribution is the *scoring formula and pipeline architecture*, not a closed-form universal $(K_R, \mathcal{S})$ rule. Table I reports a post-hoc-best row only to quantify how far the default sits from the per-dataset ceiling.

E. Stage 3 – Surrogate SHAP Attribution

For each $d \in \mathcal{R}$ we fit a LightGBM regressor [30] g_d on $(X_{\text{tr}}, s_d(X_{\text{tr}}))$, compute exact TreeSHAP [10], [11] attributions $\Phi_d^{\text{v}}, \Phi_d^{\text{te}}$, and normalise each detector by its global validation standard deviation, $\tilde{\Phi}_d^{\bullet} = \Phi_d^{\bullet}/(\text{std}(\Phi_d^{\text{v}}){+}\varepsilon)$, $\varepsilon{=}10^{-12}$. Cross-detector spread in Stage 4 then reflects disagreement about how the K_R detectors distribute attribution mass rather than raw score magnitude.

F. Stage 4 – The SHAP-σ Attribution-Evidence Gate

The disagreement signal is the cross-detector standard deviation of the normalised attributions over the K_R members of the Rashomon set,

$$\sigma_{ij} \;=\; \text{std}_{d\in\mathcal{R}}\big(\tilde{\Phi}_{d,ij}\big). \tag{4}$$

We threshold per feature: τ_j is the empirical q-quantile of $\{\sigma_{ij}^{\text{v}}\}_{i=1}^{N_{\text{v}}}$ with $q{=}0.75$, and $\phi_{ij} = \mathbf{1}[\sigma_{ij}^{\text{te}} > \tau_j]$. Marking $\phi_{ij}{=}1$ flags (i, j) as a region of *high* cross-detector attribution variance – a region where the Rashomon set, while collectively responsive, distributes attribution mass differently across detectors. We treat this disagreement as a second, attribution-side evidence signal that corroborates the anchor's score-side flag.

G. Stage 5 – Anchor Flag Generation and σ-Gated Output

The anchor produces a baseline per-point flag by a κ-sigma rule on its standardised training score, $f_i^{(a)} = \mathbf{1}[s_a(x_i^{\text{te}}) > \mu_a + \kappa\,\sigma_a]$, with $\kappa{=}3$. The canonical per-feature prediction matrix is

$$\hat{Y}_{ij} \;=\; f_i^{(a)} \wedge \phi_{ij}, \tag{5}$$

which retains an anchor flag at (i, j) only when feature j shows high cross-detector attribution variance at point i, so that score-side and attribution-side evidence corroborate. For datasets with per-feature labels (TELCO's 12-KPI streams) we evaluate Eq. (5) directly via macro F_1 across KPIs. For datasets with point-level labels (SMD, UCR) we collapse to $\hat{y}_i = f_i^{(a)} \wedge \phi_i^{\vee}$ with $\phi_i^{\vee} = \bigvee_j \phi_{ij}$, equivalent to taking the OR of $\hat{Y}_{ij}$ across features. Eq. (5) is the only fusion rule reported: no auxiliary AND/OR variants and no per-series selection that would peek at the test labels. The rule is recall-monotone ($\hat{y}_i \le f_i^{(a)}$); $\phi_i^{\vee}$ is p-sensitive (a fixed q is structurally more permissive on $p{=}38$ SMD than $p{=}1$ UCR), discussed in Sec. IV.

The complete procedure is summarised in Algorithm 1.

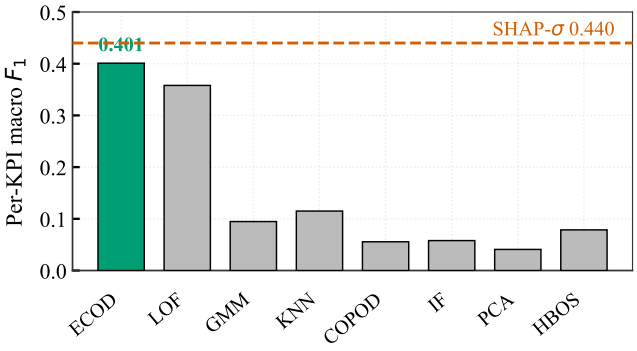


Fig. 1. TELCO per-KPI macro F_1 for each self-tuned PyOD singleton. ECOD is the best label-aware singleton (0.401); SHAP-σ exceeds it by $+0.039$ (Sec. IV-B).

III. Experiments

A. Dataset

A multivariate telemetry dataset [21] from an operational mobile cellular network spanning January–July 2021, with $p{=}12$ KPIs sampled every five minutes and ground truth labelled by Network Operation Center experts. The official test phase contains 26,496 timestamps per KPI (3,001 anomaly labels, ≈1% prevalence); [20] truncates this to 25,143 timestamps when comparing against C-MTAD-GAT.

B. Protocol

We split the training data into 80% training and 20% validation; detectors are fit on training only. Hyperparameters are self-tuned label-free on the training split (Sec. II-B). Quality q_d uses three random subspaces with 2,000 uniform background samples per subspace. The SHAP-σ quantile is fixed at $q{=}0.75$ and the anchor κ-sigma at $\kappa{=}3$.

a) Evaluation metric: We report the *per-KPI macro* F_1 together with the corresponding macro precision and recall: each metric is computed independently on each of the 12 KPI binary label streams and then averaged across streams. We compare against the released predictions of DC-VAE [21] and C-MTAD-GAT [20] on the same evaluation window.

IV. Results

A. Singleton Baselines

Figure 1 shows the standalone per-KPI macro F_1 of each label-free self-tuned detector when broadcast across the 12 KPIs. ECOD attains the highest singleton $F_1{=}0.401$, followed by LOF (0.358); the remaining six detectors stay below 0.12. No singleton matches the joint Rashomon-set selection of Section II-D.

Algorithm 1 SHAP-σ with joint Rashomon-set selection

Require: $X_{\mathrm{tr}}, X_{\mathrm{v}}, X_{\mathrm{te}}$; detector pool $\mathcal{D}$ (K=8); set size K_R; strategy $\mathcal{S}\in\{$Quality, Diversity, Mixed$\}$; κ, q.
Ensure: Predicted anomaly flags $\hat{y} \in \{0,1\}^{N_{\mathrm{te}}}$.

```
Fit every d∈D on X_tr; score all splits.
Compute MV proxy A_d via Eq. (1) and quality q_d via Eq. (2).
Compute Spearman matrix ρ_ij on {s_d(X_tr)}.
Default: if (K_R, S) unspecified, set K_R=2, S=Mixed.
Select R, |R|=K_R, by S (A: arg max q̄, B: arg max δ̄, C: arg max q̄δ̄; for K_R=2 C is Eq. (3)).
Anchor a ← arg max_{d∈R} q_d.
for all d ∈ R do
    Fit LightGBM g_d on (X_tr, s_d(X_tr)); compute TreeSHAP Φ_d^v, Φ_d^te; divide by std(Φ_d^v) + ε.
end for
Compute σ_ij via Eq. (4). τ_j ← q-quantile of σ_{·j}^v;  φ_ij ← 1[σ_ij^te > τ_j].
Compute anchor κ-sigma flag f_i^(a).
Per-feature prediction Ŷ_ij ← f_i^(a) ∧ φ_ij (Eq. (5)); for point-level evaluation return ŷ_i ← ⋁_{j=1}^p Ŷ_ij = f_i^(a) ∧ φ_i^∨.
```

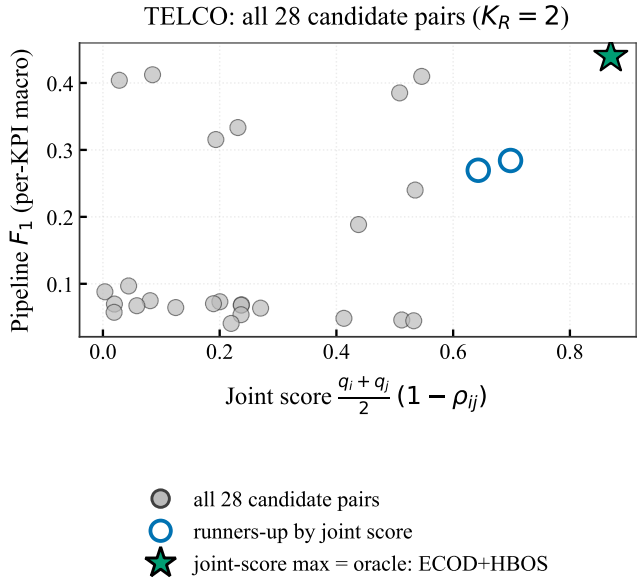


Fig. 2. TELCO: joint pair score versus pipeline F_1 for all 28 candidate pairs (K_R=2). The joint-score maximiser (green star) coincides with the empirical F_1 oracle (ECOD+HBOS); circled markers are the runners-up by joint score.

B. Joint Pair Score Identifies the Empirical Oracle

At K_R=2 the closed-form joint pair score Eq. (3) selects (ECOD, HBOS), which is exactly the label-aware empirical oracle pair under exhaustive search of all $\binom{8}{2}$=28 candidate pairs. The pair simultaneously maximises label-free quality ($\bar{q}$=0.906) and joint score (0.870). Figure 2 visualises the 28 candidate pairs: the joint-score maximiser (green star) sits at the top of the F_1 axis; the next two pairs by joint score (LOF+ECOD, LOF+HBOS) achieve mid-range F_1=0.27–0.28. With ECOD as anchor and HBOS as partner, the pipeline reaches per-KPI macro F_1=0.440, clearly exceeding the strongest singleton (0.401).

TABLE I
TELCO PER-KPI MACRO F_1, PRECISION, AND RECALL. SHAP-σ USES (K_R=2, MIXED) WITH SELF-TUNED DETECTORS; DEEP BASELINES ON THE 25,143-TIMESTAMP WINDOW OF [20].

Method	F_1	P	R
DC-VAE [21]	0.134	—	—
C-MTAD-GAT [20]	0.160	—	—
singleton ECOD	0.401	0.488	**0.396**
SHAP-σ (ECOD+HBOS)	**0.440**	**0.641**	0.376

C. Isolating the σ-filter Contribution

To separate the gain attributable to label-free *model selection* from the gain attributable to the σ *filter* itself, we compare the pipeline against the anchor singleton with the same ECOD detector. Holding the anchor fixed, the σ filter raises the per-KPI macro F_1 from 0.401 to 0.440 (*isolated* $\Delta\sigma$=+0.038). The gain is precision-driven: per-KPI macro precision rises from 0.488 (ECOD singleton) to 0.641 (+15 pts, +31% relative); recall drops only −0.020 pts, in line with the recall-monotone filter (Eq. (5)).

a) Quantile sensitivity.: Sweeping the filter quantile $q \in \{0.5, 0.75, 0.9\}$ on the ECOD+HBOS pair yields F_1 = 0.428/0.440/0.404 with precision 0.58/0.64/0.62: q=0.75 optimises F_1 across the three operating points, confirming robustness within a $\sim$ 0.04 envelope.

D. Comparison to Deep Baselines

On the 25,143-timestamp truncated window of [20], DC-VAE [21] attains per-KPI macro F_1=0.134 and C-MTAD-GAT [20] F_1=0.160; SHAP-σ reaches F_1=0.446, a 3.3–2.8× margin over the two deep baselines on the same evaluation window. Table I summarises the headline comparison.

V. CONCLUSIONS

We presented SHAP-σ, a label-free pipeline for multivariate cellular network anomaly detection that promotes the cross-detector standard deviation of SHAP attributions from a post-hoc explanation into a first-class attribution-evidence signal.

The methodological contributions are: (i) label-free self-tuning of every PyOD detector (PCA, k-NN, LOF, GMM, HBOS) from training-data statistics; (ii) a closed-form Rashomon pair score $\frac{q_i+q_j}{2}(1-\rho_{ij})$ combining a Mass–Volume proxy with a cardinality veto and pairwise Spearman diversity; and (iii) a per-KPI σ gate that filters anchor flags by the cross-detector spread of TreeSHAP attributions. On the TELCO cellular benchmark the closed-form pair score selects the empirical oracle pair (ECOD+HBOS) label-free; the pipeline reaches per-KPI macro $F_1{=}0.440$ ($+0.039$ over the best singleton, with isolated σ-contribution $\Delta\sigma{=}+0.038$), and exceeds the released deep baselines DC-VAE and C-MTAD-GAT [20], [21] by 2.8–3.3$\times$ on the same evaluation window.

a) Limitations and future work.: Eq. (5) is recall-monotone with respect to the anchor singleton; the disjunctive feature-wise gate is most informative on per-feature labels (as in TELCO) and degenerates under OR-aggregation for point-level multivariate evaluation, which we defer to a journal extension on extended cellular benchmarks. We also plan to learn $(K_R, \mathcal{S})$ label-free from synthetic anomaly injection [6], [8], to compare against recent deep label-free baselines such as MTGFlow [**?**], and to extend σ-SHAP to cross-window variance for streaming deployment.

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